Junu Lee

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Education

The Wharton School, University of Pennsylvania — Philadelphia, PA
Ph.D. in Statistics and Data Science — 08/2022 – 05/2027 (*expected*)
Advisors: Zhimei Ren, T. Tony Cai

Harvard University — Cambridge, MA
A.B. in Mathematics and Statistics — 08/2018 – 05/2022
S.M. in Computer Science (concurrent AB/SM program)
Advisor: Lucas Janson

Research interests

Multiple hypothesis testing; Selective inference; e-values; Distribution-free inference; Conformal inference; Agentic data science; Data-driven decision making

Papers and preprints

* *denotes alphabetical order or equal contribution*

[7] **Junu Lee**, Rina Foygel Barber, Richard J. Samworth, and Zhimei Ren (2026). Compound p-values for competition-based testing. arXiv preprint arXiv:2610.20411.

[6] Abhinav Chakraborty*, **Junu Lee***, and Eugene Katsevich (2026). Power of masking methods for adaptive testing in a multivariate normal means problem. arXiv preprint arXiv:2601.07764. Under review at *Biometrika*.

[5] **Junu Lee**, Ilia Popov, and Zhimei Ren (2025 +). Full-conformal novelty detection. arXiv preprint arXiv:2501.02703. Major revision requested at *Journal of the American Statistical Association*.

[4] Jeffrey Zhang and **Junu Lee** (2025). A general condition for bias attenuation by a nondifferentially mismeasured confounder. *Biometrika*, 112(3), asaf026.

[3] **Junu Lee** and Zhimei Ren (2024+). Boosting e-BH via conditional calibration. arXiv preprint arXiv:2404.17562. Minor revision requested at *Journal of the Royal Statistical Society: Series B*.

[2] Taehyeon Kim, Eric Lin, **Junu Lee**, Christian Lau, and Vaikkunth Mugunthan (2023). Navigating data heterogeneity in federated learning: A semi-supervised federated object detection. *Advances in Neural Information Processing Systems*, 36, 2074–2096.

[1] Eugene Curtin*, **Junu Lee***, Andrew Lu*, and Sophia Sun* (2019). A modified Grassmann algebra approach to theorems on permanents and determinants. *Linear Algebra and its Applications*, 581, 20–35.

Software

1. `fcnd`: Python package for full conformal novelty detection methods. [GitHub] [PyPI]

Awards and honors

J. Parker Memorial Bursk Prize for Excellence in Research 2026
University of Pennsylvania, Department of Statistics and Data Science

Rising Star in Data Science 2026
University of Chicago, Harvard University, UC San Diego, Stanford University

Rising Star in Statistics and Data Science 2026
Cornell University

IMS Hannan Graduate Student Travel Award 2026
Awarded for the 2026 Joint Statistical Meetings

Lawrence D. Brown Student Paper Award Symposium Speaker 2024, 2025
University of Pennsylvania, Department of Statistics and Data Science

NSF Graduate Research Fellowship 2022

Harvard College Research Program 2020

Herchel Smith Harvard Undergraduate Research Fellowship 2020

Derek Bok Certificate of Distinction in Teaching 2019, 2020
Harvard University

Talks

Invited talks:

10. **Compound p-values for competition-based testing**
    Rising Stars in Data Science Workshop
    University of Chicago, November 2026 (upcoming)
9. **Compound p-values for competition-based testing**
    Rising Stars in Statistics and Data Science Workshop
    Cornell University, September 2026
8. **Power of masking methods for adaptive testing in a multivariate normal means problem**
    International Seminar on Selective Inference
    Virtual, April 2026
7. **Boosting e-BH via conditional calibration**
    Emmanuel Candès' Group Meeting
    Stanford University, March 2026
6. **Full-conformal outlier detection**
    Lawrence D. Brown Student Paper Award Symposium
    University of Pennsylvania, October 2025
5. **Full-conformal outlier detection**
    Multiple Comparison Procedures (MCP)
    Philadelphia, August 2025
4. **Boosting e-BH via conditional calibration**
    BIRS Invited Workshop: Game-theoretic statistical inference
    Chennai, June 2025

3. **Using e-values for conformal multiple testing**
   Wharton Statistics First Year Ph.D. Student Seminar
   University of Pennsylvania, April 2025
2. **Boosting e-BH via conditional calibration**
   Lawrence D. Brown Student Paper Award Symposium
   University of Pennsylvania, November 2024
1. **Boosting e-BH via conditional calibration**
   International Seminar on Selective Inference
   Virtual, June 2024

Contributed talks:

4. **Competition-based testing with compound p-values and e-values**
   Joint Statistical Meetings (JSM)
   Boston, August 2026
3. **Boosting e-BH via conditional calibration**
   Joint Statistical Meetings (JSM)
   Nashville, August 2025
2. **Boosting e-BH via conditional calibration**
   New England Statistical Symposium (NESS)
   New Haven (given virtually), June 2025
1. **Boosting e-BH via conditional calibration (with applications to variable selection)**
   Eastern North American Region (ENAR)
   New Orleans, March 2025

Teaching

Certifications:

Certificate in College and University Teaching
Awarded by Penn's Center for Excellence in Teaching, Learning, and Innovation

Teaching Assistant, The Wharton School

STAT 9270*‡: Bayesian Statistical Theory and Methods	Spring 2024
STAT 4050‡/7050†‡: Statistical Computing with R	Fall 2023 (Q2)
STAT 1110: Introductory Statistics	Fall 2023
STAT 1020: Introductory Business Statistics	Fall 2022

Teaching Assistant, Harvard University

STAT 149: Introduction to Generalized Linear Models	Spring 2022
STAT 210*: Probability I	Fall 2021
STAT 111: Introduction to Statistical Inference	Spring 2020, 2021
STAT 110: Introduction to Probability	Fall 2019, 2020, 2021

**: graduate-level courses; †: MBA courses; ‡: guest lectures given*

Training

Undergraduate Student Research Supervision

- Ilia Popov (2024-2025); *next position: PhD Student, Stanford University, Department of Statistics.*

Professional service

Wharton Directed Reading Program January 2026 – April 2026

Organized and designed a directed reading curriculum on multiple hypothesis testing to undergraduate students.

Wharton Doctoral Program Ph.D. Mentor August 2024 – *Present*

Provided mentorship to incoming Ph.D. students in the Wharton doctoral programs.

Invited Discussions:

International Seminar on Selective Inference:

- "E-statistics, group invariance and anytime-valid testing" by Muriel Pérez-Ortiz, December 2024
- "Bringing Closure to FDR Control With a Uniform Improvement of the e-Benjamini-Hochberg Procedure" by Neil Xu, May 2025

Conferences:

- Chair of the invited session "Conformal inference and statistical testing for reliable deployment of AI/ML models" at Joint Statistical Meetings (JSM), 2025
- Organizer and co-chair of the invited session "The role of e-values in multiple testing" at Multiple Comparison Procedures (MCP), 2025

Journal Reviewer: Journal of the American Statistical Association; Biometrika; Communications in Statistics - Theory and Methods

Industry experience

Dynamo AI *Various*

Founding member, advisor 05/2022 – 12/2023

- Developed a multitude of federated learning algorithms and adapted them to various problem domains, e.g., finance, automated driving.
- Met with industry partners to collaborate on algorithms and present finished products.

Hudson River Trading New York, NY

Algorithm developer intern 06/2021 – 08/2021

- Completed multiple quantitative projects on financial forecasting.

Proton.ai Boston, MA

ML research intern 05/2019 – 08/2019

- Developed NLP-based recommendation systems for sales teams to efficiently market new products and promotions.
- Contributed to the research and training of customer churn models.